Random Forest Cancer Classification Using The Cancer Genome Atlas (TCGA)

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Prediction Score: 1.00 | Ranking: 4/10
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1. Problem Statement

Cancer is a disease of genome alterations: DNA sequence changes, copy number aberrations, chromosomal rearrangements and modification in DNA methylation together drive the development and progression of human malignancies (Brennan et al., 2013). With the complete sequencing of the human genome and continuing improvement of high-throughput genomic technologies, it is now feasible to contemplate comprehensive surveys of human cancer genomes. The Cancer Genome Atlas (TCGA) aims to catalogue and discover major cancer-causing genome alterations in large cohorts of human tumours through integrated multi-dimensional analyses.

Using the TCGA, this project addresses two objectives. The first is to accurately classify three types of cancer: squamous cell carcinoma (LUSC), ovarian cancer (OV), and glioblastoma multiforme (GBM), based on log-space gene expression measurements across 12,042 genes from 886 patient samples. The second is to identify and assess potentially mislabeled observations in the training data, as real-world genomic datasets are subject to labeling errors that can systematically degrade model performance, particularly for underrepresented classes. These two objectives are closely related, since accurate mislabel detection is a prerequisite for reliable classification, especially given the class imbalance in our dataset (GBM=408, OV=348, LUSC=130), where label noise may disproportionately affect the minority class.

The three cancer types targeted in this study represent serious and distinct clinical challenges. Squamous cell lung cancer, also known as squamous cell carcinoma of the lung, accounts for approximately 30% of all lung cancers (*Squamous Cell Lung Cancer*, 2024). It is the second most common subtype of non-small-cell lung cancer and leads to 50,000 deaths per year in the United States (Liao et al., 2012). Equally important is ovarian cancer, which is diagnosed in an estimated 300,000 people annually and contributes to approximately 180,000 deaths worldwide (Zamarin, 2025). Glioblastoma multiforme is a very rare but highly fatal brain tumor, affecting 4 out of 100,000 people in Canada (*Glioblastoma (GB)*, 2019). These statistics emphasize the importance of improving current methods for early and accurate cancer detection.

Despite advances in treatment, multiple cancers, particularly GBM and OV, are difficult to diagnose at an early stage when interventions prove to be the most effective (Gao et al., 2020). Leveraging genome alterations and machine learning techniques, such as discriminant analysis, random forests, or boosting, offers the potential to identify patterns in genetic data that may not be apparent through traditional diagnostic approaches. By accurately classifying cancers such as LUSC, OV, and GBM based on genomic signatures, this research could contribute to faster and more precise diagnostic tools. Ultimately, identifying the genes most strongly associated with these cancers may support earlier detection, improve personalized treatment strategies, and enhance clinical decision-making, helping reduce mortality and improve patient outcomes.

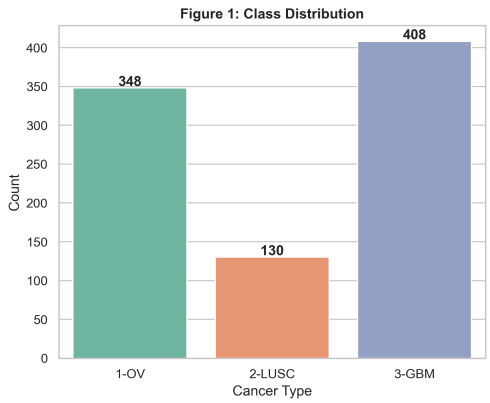
Research between these types of cancer and related genes is limited but emerging. TOP2A, CCNA2, CDC20, AURKA, AURKB, and FENI, have been identified as key genes related to the progression of LUSC, and in

more than 96% of patients with OV, mutations in the single gene TP53 are commonly present (Gao et al., 2020; Bethesda, 2011). Note that TP53 encodes a tumor suppressor protein that normally prevents cancer formation. Mutations in the gene disrupt this protein's function, which contributes to uncontrolled growth of ovarian cells (Bethesda, 2011). In addition, TCGA identified low prevalence but statistically recurrent somatic mutations in 9 additional genes including NF1, BRCA1, BRCA2, and CDK12 (Gao et al., 2020).

2. Methods

2.1. Data Preprocessing & EDA

The dataset, obtained from TCGA, contains 886 observations and 12044 features. The first feature is a sample identifier, which serves to distinguish observations while preserving patient anonymity. The second feature is the outcome variable, cancer type, encoded as a categorical variable with the following three classes: 1 = Glioblastoma Multiforme (GBM), 2 = Lung Squamous Cell Carcinoma (LUSC), and 3 = Ovarian Cancer (OV). The remaining 12,042 features represent individual gene expression measurements in log space. These gene expression values reflect the activity levels of thousands of genes. For example, TP53, a well-known tumor suppressor gene, or EGFR, a gene frequently implicated in GBM and LUSC. Note that the dataset contains no missing values across any of the features.



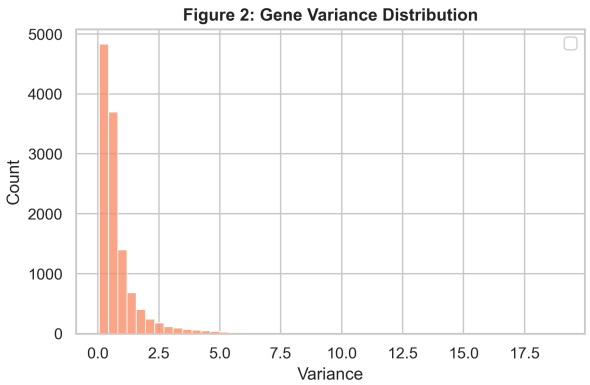
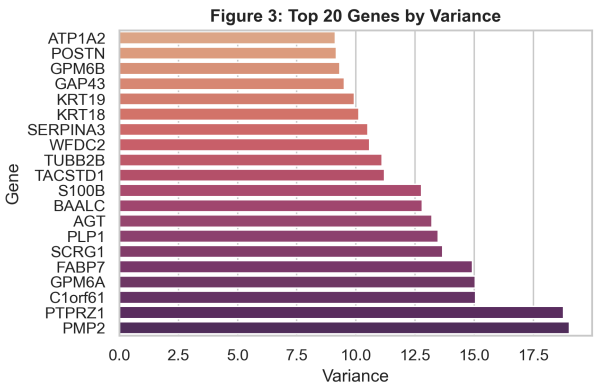


Figure 1 shows that the class distribution across cancer types in the data set is notably imbalanced. GBM accounts for the largest share of observations with 408 samples (46.1%), followed by OV with 348 samples (39.3%), and LUSC with the smallest representation at 130 samples (14.7%). Figure 2 shows the variance distribution for the genes in the dataset: the cluster of genes near zero (approximately 4700) suggests that the majority of genes are not very informative given that they barely change across observations.



Lastly, Figure 3 showcases the genes with the highest variance. In other words, the most informative genes as they vary the most across cancer types. PMP2 and PTPRZ1 stand out as by far the most variable. Interestingly, several of these are biologically meaningful: PTPRZ1 is a known GBM marker (Huan et. al., 2021), and PLP1, FABP7, SCRG1 are all associated with brain/neural tissue (Wolf, van Spaendok & Hobson, 2025; Kaloshi et al, 2007; Wang et al., 2019), which makes sense given GBM is a brain cancer. Additionally, KRT18, KRT19, WFDC2 are associated with epithelial/ovarian tissue, which aligns with OV (Sun et al., 2023; Moskov et al. 2024).

To fulfill our objective of classifying cancer subtypes from high-dimensional gene expression data using supervised machine learning methods, preprocessing and dimensionality reduction were required prior to model fitting, given the large number of predictors relative to observations (p >> n). Data preprocessing began by removing non-informative variables and splitting the data into training and validation sets for parameter tuning and model assessment. The response variable (cancer type) was separated from the predictors, and the sample identifier (id) was excluded. The dataset was then split into training (n=708) and validation (n=178) sets using stratified random sampling to preserve class proportions in these sets, ensuring that model training and evaluation were not biased by class imbalance.

To reduce dimensionality, gene-level variance was computed within the training set and used as a filtering criterion. This method, known as *Variance Filtering*, is considered advantageous in high-dimensional gene expression data sets as it can help improve predictive performance, run time and interpretability by removing irrelevant and redundant features before training a model (Bommert et. al, 2022). Low-variance genes typically contribute little to class discrimination, while highly variable genes are more likely to capture meaningful differences between cancer types. Based on the scree plot above, the top 1000 most variable genes were retained for further analysis; notice that after 1000, most genes have variance smaller than 2.5. This selection was performed using only the training data, and the same subset of genes was applied to the validation set to avoid data leakage.

2.2. Pipeline A: Variance Filtering + Random Forest

After the variance filtering step, we proceeded with model fitting. The primary model used was a *Random Forest* classifier, chosen for its strong performance in high-dimensional settings and its ability to capture complex, non-linear relationships without strong distributional assumptions. Random forests reduce variance through aggregation of many decision trees, each fitted on a different bootstrap sample, and introduce additional randomness by selecting a random subset of m < p predictors at each split, which decreases correlation between trees and improves generalization over bagging.

The model was trained using B = 1000 trees, a value supported by prior studies in genomic classification tasks showing diminishing returns in error reduction beyond this value (Diaz-Uriarte & Alvarez de Andres, 2006). Two hyperparameters were tuned: the number of predictors considered at each split (*m*) and the minimum number of observations per terminal node (*node_size*), which controls how deep the trees are allowed to grow. A grid of candidate values was evaluated, ranging from small subsets of predictors to the full set {10, 20, 30, 50, 75, 100, 150, √p, p/2, p}, as well as varying levels of nodesize {1, 3, 5, 10}. To address class imbalance, class weights were set inversely proportional to class frequencies, penalizing misclassification of the minority class (LUSC) more heavily during training. Parameter selection was based on out-of-bag (OOB) accuracy, an internal validation approach that estimates performance using observations not included in the bootstrap samples. Since each tree is trained on approximately 63% of the data (the ones that are bootstrapped), the remaining 37% of samples can be used as "training data" since the model never sees them during training, providing unbiased probability estimates.

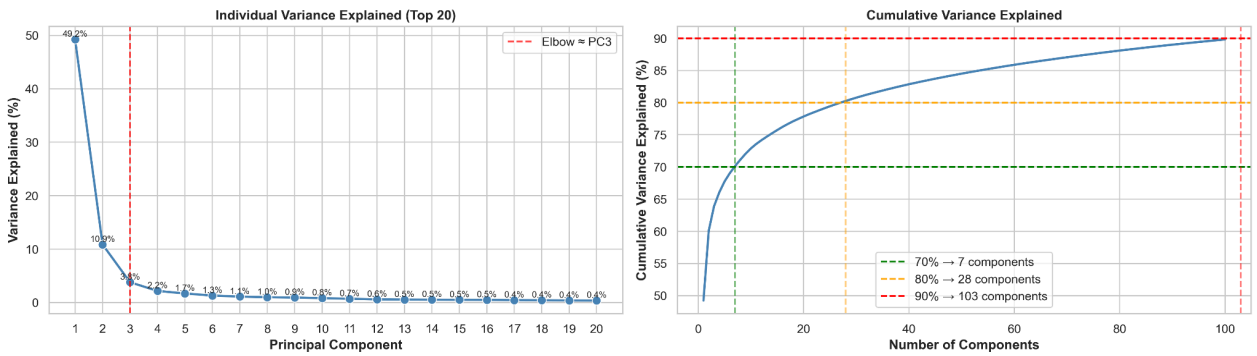
The optimal parameters were m = 75 and nodesize = 1 (fully grown trees), achieving an OOB accuracy of 89.55%. Examining class-specific performance revealed a significant disparity: OV and GBM achieved over 94% correct classification in the OOB samples, while approximately 36% of LUSC observations were misclassified. This confirmed LUSC as the primary classification challenge, motivating the mislabeled observation analysis described in Section 2.5.

2.3. Pipeline B: PCA + LDA

Next, we attempted another commonly used machine learning method in cancer classification problems: Linear Discriminant Analysis combined with Principal Component Analysis (Wei et. al., 2023; Sharma & Paliwal, 2008, Esen et al., 2024). Although we considered regularized LDA to reduce dimensionality, we decided to proceed with PCA because it is better suited for really high-dimensional data.

Some advantages of LDA is that it estimates a probability function for each class and sets interpretable linear boundaries between classes. We believe this may be useful in detecting mislabeled observations, as we can compare the probability of a label being in the correct class and its true label: if the probability of being in its true label is sufficiently small, we may label the observation as suspicious. We attempt this in Section 2.5. Note that this model was fit under the assumption that observations come from different Gaussian distributions (represented by clusters), and that variance is the same across classes in LDA. We also had to standardize the gene expression to have mean = 0 and sd = 1 before performing PCA.

Figure 5: PCA Scree Plots



The Scree plot to the right shows that PC1 (principal component 1) explains 49.3% of the variance alone. After PC3-4 it drops to nearly zero and becomes flat. Combining this with the cumulative variance plot to the right, we tuned the n = *number_of_PCs* for the LDA from the following grid: {1, 2, 7, 15, 20, 28, 50}. Although performance peaked at n=2 components for overall validation accuracy (90.45%), n=7 components was selected as the final choice as it captures 70% of cumulative variance (consistent the cumulative variance graph), and achieved a higher LUSC recall (61.54%) than n=2 (57.61%), even though it has slightly lower validation accuracy (89.89%).

2.4. Pipeline C: Boosting

Following the results from PCA+LDA and Random Forest, we implemented Gradient Boosting as a third ensemble method, aiming to further improve classification accuracy. Unlike Random Forest, where trees are grown independently in parallel, Gradient Boosting builds trees sequentially following Algorithm 8.2 from James et al. (2021). At each iteration b = 1, ..., B, a new tree is fitted to the residuals of the current model, and the ensemble is updated as using λ, the learning rate that controls the contribution of each tree. The final model is the sum of all B trees with the learning rates.

The three key tuning parameters are B (number of trees), λ (learning rate/shrinkage), and d (tree depth). A grid search over $\lambda \in \{0.001, 0.01, 0.1, 0.2\}$ and $d \in \{1, 2, 3, 4\}$ was conducted with B fixed at 1000 for computational efficiency. The optimal parameters were $\lambda = 0.1$ and $d = 2$. Notably, $\lambda = 0.01$ yielded identical validation accuracy as $\lambda = 0.1$, but was excluded in favour of $\lambda = 0.1$ for computational efficiency. Since $\lambda = 0.1$ was selected, B = 500–1000 trees is appropriate per ISL (James et al., 2021), and B = 1000 was confirmed by Díaz-Uriarte & Alvarez de Andrés (2006).

Due to the computational cost of sklearn's GradientBoostingClassifier, we switched to XGBoost (Chen & Guestrin, 2016), an optimized implementation that adds L1/L2 regularization and parallelization while following the same core algorithm. All tuned parameters were kept identical across both implementations.

2.5. Mislabeled Observations

We employed two independent approaches to identify potentially mislabeled observations: Cleanlab and PCA+LDA, allowing us to cross-validate our findings across fundamentally different methodologies.

The first approach uses Cleanlab, a Python library that implements a technique called Confident Learning (Northcutt et al., 2021; cleanlab, 2026)). Rather than relying on the model's training predictions, which would be overconfident and circular, Cleanlab leverages the Out-of-Bag (OOB) probability estimates from the Random Forest. A sample is flagged as suspicious when the model consistently and confidently assigns a higher probability to a different class than the one given; for example, a sample labeled as LUSC that the model repeatedly classifies as GBM with high confidence is labeled suspicious.

Applied to the training set (n=708), Cleanlab flagged 70 suspicious observations. The most striking finding was in LUSC: 13 samples labeled as LUSC were suspected to actually be GBM, and 21 were suspected to actually be OV, meaning 34 out of 104 LUSC samples (32.7%) were potentially mislabeled, explaining the low recall observed previously. By comparison, 19 GBM and 17 OV samples were flagged, but given their larger sample sizes, the impact of label noise is far less severe for these classes. An additional 11 suspicious observations were identified in the validation set, bringing the total to 81 flagged observations out of 886 (9.1%), consistent with the instructor's note that a small proportion of labels were deliberately flipped.

The second approach used the PCA + LDA model we fitted in Section 2.3., to assign class membership probabilities to each sample. A sample is flagged as suspicious if the model assigns a probability lower than 0.33, the random chance baseline for three classes, to its given label. For example, if a sample is labeled as LUSC, but the P(LUSC) is less than 0.33, we label it as suspicious. Applied to the training set, with this threshold, this approach flagged 92 suspicious observations (13.0%).

Strikingly, all 70 samples flagged by Cleanlab were also identified by PCA+LDA, providing strong independent confirmation that these observations are likely mislabeled. The 22 additional samples flagged only by PCA+LDA likely represent borderline cases. The agreement between two different approaches, one based on Random Forest OOB probabilities and the other on a probabilistic linear discriminant model, substantially strengthens our confidence in the mislabeled sample identification.

Next, we used these findings to see if we could improve model accuracy. We chose to remove rather than relabel suspicious observations, as relabeling would introduce a new source of uncertainty since we cannot be certain the suggested labels are correct, and we would rather lose 81 out of 886 observations than introduce 81 unreliable ones.

3. Results

3.1. Mislabeled Observations

Cleanlab and PCA+LDA had a 100% overlap in mislabeled observations, with PCA+LDA flagging a few additional ones. After removing the 70 flagged training samples and refitting the Random Forest, OOB accuracy improved dramatically from 89.55% to 99.22%, and validation accuracy (on the cleaned set without suspicious observations) improved from 91% to 97%, with LUSC recall improving significantly. However, we believe these results could be misleading: the 99.22% OOB accuracy is suspiciously high and likely reflects an oversimplified training set rather than a genuine improvement in prediction.

To obtain a more honest estimate of true performance, we evaluated the cleaned model on the full validation set including suspicious observations. This yielded identical results to the original model fitted on the full training data (91% accuracy, 62% LUSC recall), suggesting two important conclusions. First, Random Forest is inherently robust to label noise due to its ensemble averaging so there may be no need to exclude these observations in the final model; and second, the apparent improvement when evaluating on the cleaned validation set was driven by the removal of hard-to-classify samples in the validation set rather than a genuine improvement in model performance.

Metric	Before Cleaning	Cleaned training set + Cleaned validation set	Cleaned training set + Full validation set
OOB Accuracy	89.55%	99.22%	99.22%
Validation	91.01%	97.01%	91.01%
LUSC Recall	61.54%	80%	61.54%

Table 1: Comparing Random Forest Results with vs. without suspicious observations

As a result, we decided not to remove any suspicious observations to train our final model, since improvement appears to be just an oversimplification of the training and validation set, that may not generalize to the final test set. This might also suggest that ensemble tree models like Random Forest are already highly robust to mislabeled observations.

3.2. Model Comparisons

Metric	Random Forest	PCA+LDA (n=7)	XG Boost
Validation Accuracy	91.01%	89.89%	92.13%
LUSC Recall	61.54%	61.54%	65.38%
OV Recall	97.14%	94.29%	97.14%
GBM Recall	95.12%	95.12%	95.12%

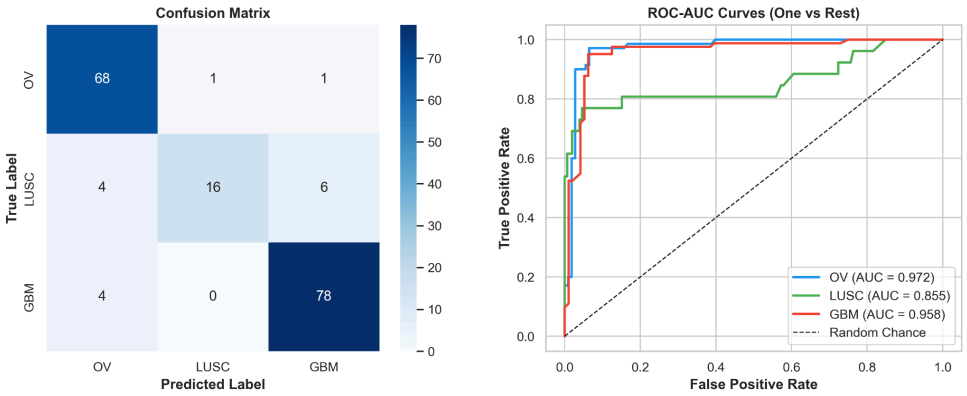
Table 2: Model Comparisons

Note that all models were fitted without removing any suspicious observations, either from the training or validation sets. Interestingly Boosting outperformed Random Forest on the validation set, and also achieved

the highest LUSC recall (65.38%). Note that the improvement in LUSC recall was only modest, calling for the need to collect more data on this category or a possibility of mislabeled observations. The results also suggest that ensemble tree methods tend to outperform methods that impose linear boundaries, like PCA+LDA, when working with high-dimensional data. Note that PCA+LDA was still surprisingly close to Random Forest, despite being considerably simpler.

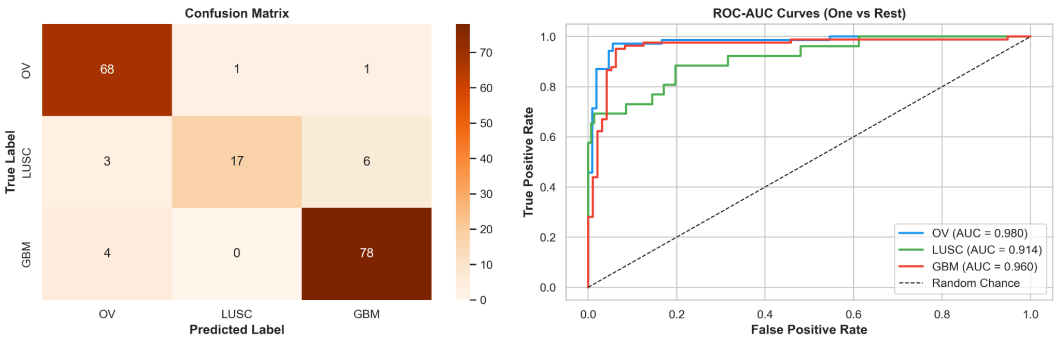
Next, we compare the two top performing models, Random Forest and Boosting, on the validation set.

Figure 6: Random Forest Model Evaluation



Random Forest obtained a 91% accuracy on the validation set, which is good given the class imbalance. Both OV and GBM achieved very high Recall and Precision rates, and also very high AUC. Out of 70 OV samples only 2 were misclassified, and out of 82 GBM samples, only 4 were misclassified. On the other hand, out of 26 LUSC samples, 10 were misclassified. Interestingly, LUSC achieved a very high precision (94%), but a very low Recall (62%), meaning that the model was good at getting little false positives, but it missed real LUSC cases very often.

Figure 7: XGBoost Model Evaluation



The most outstanding finding from Figure 6 is that the ROC curve for LUSC looks much better for XG Boost than for Random Forest ($AUC_{BOOST} = 0.91 > AUC_{RF} = 0.85$), contributing to the higher accuracy of XG Boost in the validation set. We observe identical accuracies for GBM and OV classes, the only difference is that XG Boost classified one additional LUSC sample correctly, compared to Random Forest. Given the small size of this class, this explains the significant improvement in AUC.

Finally, observe the Kaggle results in Table 3 below. Even though Boosting outperformed Random Forest in the validation set, Random Forest outperformed Boosting in the Kaggle submission. Hence, we decided to select Random Forest as our final model. The table also shows that fitting the Random Forest with or without suspicious observations yielded the same test accuracy, suggesting the Random Forest is highly robust to noisy or mislabeled data points. Additionally, we believe our XG Boost model might have been overfit, because it failed to generalize to the test data, though it performed well on the training and validation sets.

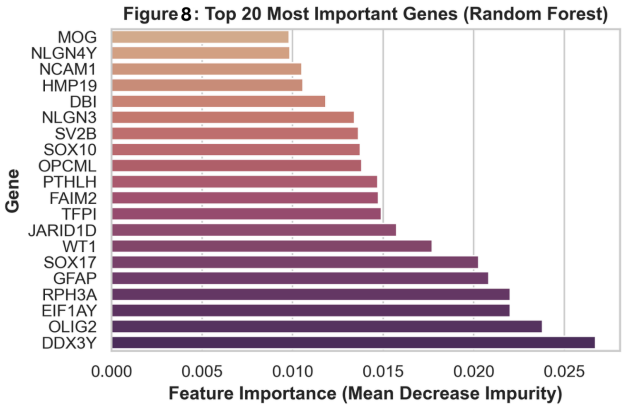
Metric	Random Forest	Random Forest clean	PCA+LDA (n=7)	XG Boost
Test Accuracy	100%	100%	97.98%	98.99%

Table 3: Final Kaggle Results

3.3. Final Remarks and Takeaways

Aside from the prediction model for future use, Random Forest offers an interpretable output as well. We can observe which features lead to the greatest decrease in mean impurity in Figure 8, in other words, which features were able to better distinguish between the classes. Note that most of these are different from the ones we discussed as most variables, suggesting that although variability is important in feature selection, it is not the only thing we should consider.

One of the key insights from this plot is that top genes are heavily related to brain/neural tissue, which makes sense given GBM is a brain cancer, for example, OLIG2, GFAP, NCAM1, and NLGN3 (Szu et al., 2023). These are related to important brain functions like neural cell adhesion or transcription factors in neural cell development. We also observe genes linked to the Y-chromosome, like DDX3Y and EIF1AY, which capture the sex-based differences in GBM incidence. Since men tend to be diagnosed with glioblastoma more often than women, having a Y chromosome and certain gene expression can easily indicate a sample is GBM (Sun et al, 2014). Moreover, genes like WT1, PTHLH and OPCML are strongly associated with ovarian cancer progression, also observed in Figure 8 (Shao et al., 2021). Lastly, it is important to mention that almost none of the top 20 genes are LUSC-specific; the model is good at distinguishing GBM from OV, and likely LUSC falls out as a residual category.



We'd also like to mention a few of the takeaways about building machine learning pipelines we learned from this project. First, it is important to always consider computational costs and run time when deciding how to tune parameters. Basing ourselves on literature was of great help in tuning parameters because at times, our code would run for over 20 minutes without showing clear results. Although we understand each dataset is different, we believe literature can be a good starting point in deciding which parameter values to consider. Second, we learned the importance of checking that test and train data have the same labels before getting started, to ensure smooth prediction later on. And third, we learned that when working with multiple classes, it is key to check for class imbalances and ensure random samples are drawn accordingly; for example implement balance in the random forest and in the training/validation set split.

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5. LLM / GPT Policy

This project used Claude AI and GPT-4 to brainstorm the initial project outline. Particularly, a timeline addressing focal points of the project. For example, during the first week, conduct exploratory data analysis to verify missing values and spread of predictor variables. Then, during the second week, try out different machine learning models (including linear discriminant analysis, decision trees, random forests, and boosting). During the third week, choose a model and start drafting the initial report, and so on. This allowed our team to have a clear notion of what needed to be completed by each passing week. Using this timeline, our team remained organized and was able to explore the concepts that the artificial intelligence platforms suggested. Although this was extremely useful, it only served as an initial guideline for our project. We complemented their suggestions with information from the lecture slides, the textbook, our own research, and office hours conversations with both the teaching assistants and the professor. This way, we ensured that the final decisions, code, and writing, were entirely produced by our team.

6. Code

The Jupyter Notebook containing our code can be found in the next pages.